Performance Testing

OptiCrop — Smart Agricultural Production Optimization Engine

Methodology

The dataset (2,200 samples, 22 balanced crop classes) was split 80/20 into training and test sets using stratified sampling, ensuring every crop class was proportionally represented in both sets. Four classifiers were trained on the same training split and evaluated on the same held-out test split for a fair comparison.

Model Comparison Results

Generalization Check

5-fold cross-validation was additionally run during model selection to confirm the top-performing model (Random Forest) was not simply overfitting to one particular train/test split.

Application-Level Testing

- Verified known input combinations (e.g. high rainfall + high humidity) return agronomically sensible crops (e.g. rice).
- Tested edge cases: empty/non-numeric form fields correctly return an error message instead of crashing.
- Confirmed all 3 application pages (Home, About, Find Your Crop) load correctly and navigation between them works.
- Re-loaded the saved model/scaler/encoder from disk in a fresh session and confirmed identical predictions to the in-memory trained versions — validating the serialization process.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest (selected)	99.55%	99.57%	99.55%	99.55%
KNN	97.95%	98.04%	97.95%	97.93%
Decision Tree	97.95%	98.06%	97.95%	97.94%
Logistic Regression	97.27%	97.40%	97.27%	97.25%